Conversion and Validation of TeamNando Distribution Network Models from OpenDSS to Pandapower

A. Overview

This work focuses on the conversion of detailed distribution network models from OpenDSS to pandapower, aiming to enable a consistent comparison between the two simulation environments. The modelling process includes the representation of network topology, electrical parameters, and control elements such as transformers, regulators, and capacitors. Following the conversion, time-series power flow simulations are performed and key results, such as voltage profiles and loading conditions, are analysed and compared in order to evaluate the consistency, accuracy, and limitations of the two tools.

Description of the Studied Distribution Networks

The study focuses on detailed modelling of real distribution networks provided by AusNet Services, consisting of integrated high-voltage (22 kV). Four representative feeders were selected, including both rural and urban configurations, in order to capture a wide range of network characteristics such as feeder length, number of customers, loading conditions, and PV penetration levels. These networks include various components such as distribution transformers, voltage regulators, capacitors, and single-wire earth return (SWER) systems, particularly in rural areas. Due to the lack of detailed LV data, the low-voltage networks were modelled using realistic assumptions based on standard design practices. This effectively results in an aggregated representation of LV demand at the transformer level, which does not fully capture feeder topology and phase imbalance, as discussed in [1]. Additionally, realistic time-series demand profiles derived from smart meter data were used to simulate customer consumption behaviour, enabling a more accurate representation of network operation and performance under varying load conditions.

OpenDSS

OpenDSS is an open-source power system simulation tool developed by the Electric Power Research Institute (EPRI), widely used for distribution system analysis. It supports detailed modelling of unbalanced, multi-phase networks and is particularly suitable for time-series simulations, enabling the study of dynamic behaviours such as load variations and distributed generation integration.

Pandapower

Pandapower is an open-source Python-based power system analysis tool that builds on the PYPOWER framework, offering a user-friendly and flexible environment for modelling and simulation. It enables efficient power flow calculations, time-series analysis, and integration with data science libraries, making it well-suited for automated studies, optimization tasks, and comparison with other simulation tools.

B. Implementation

The developed framework follows a pipeline-based approach for the conversion and validation of distribution network models from OpenDSS to pandapower. The process includes data extraction, model transformation, time-series simulation, and result comparison against OpenDSS reference

outputs. The implementation is modular, allowing each stage of the pipeline to be executed independently or as part of an automated workflow.

Pipeline Description

The pipeline consists of the following main stages:

- **Data Preparation:** Generation of OpenDSS input files from structured Excel network data.
- **Model Conversion:** Transformation of the MV and LV network elements into pandapower format.
- **3-Phase Preparation:** Extension of the network to support unbalanced three-phase analysis.
- **Simulation:** Execution of time-series power flow simulations in both balanced and unbalanced modes.
- **Validation:** Comparison of pandapower results against OpenDSS simulations using voltage and loading metrics.

Data Preparation

The first stage of the pipeline involves the generation of OpenDSS input files from structured Excel datasets describing the network topology and components. These datasets include information such as buses, lines, transformers, loads, and network parameters. A dedicated script processes the Excel files and converts them into a set of OpenDSS (.dss) files, ensuring that all required elements (e.g., line codes, load shapes, and control devices) are properly defined. This step enables a standardized and reproducible representation of the network in OpenDSS format.

Model Conversion

The medium-voltage (MV) network is then constructed in pandapower by parsing the corresponding OpenDSS files. The conversion process includes the extraction of network topology, line parameters, and bus connections, which are mapped into pandapower elements such as buses and lines. Special attention is given to preserving electrical characteristics, including impedance values and connectivity, to ensure consistency between the original OpenDSS model and the converted pandapower representation.

LV Network Integration

Following the MV conversion, the low-voltage (LV) network is incorporated into the pandapower model. This includes the addition of distribution transformers, LV feeders, and connected loads. Since LV data may not always be explicitly available, assumptions based on standard design practices are used to approximate the LV topology and customer distribution. Such approximations may introduce deviations, as simplified representations of LV networks do not fully capture their unbalanced and radial characteristics [1].

3-Phase Model Preparation

To enable unbalanced analysis, the network is extended to a three-phase representation. This includes the incorporation of zero-sequence parameters and the modelling of asymmetric loads across phases. The preparation step ensures compatibility with three-phase power flow solvers and allows the study of phase imbalance effects, which are particularly important in distribution networks.

Time-Series Simulation (pandapower)

Time-series power flow simulations are performed using pandapower to analyse the dynamic behaviour of the network. Simulations are conducted over a 24-hour period with a 30-minute resolution, capturing variations in load demand. Both balanced and unbalanced simulations are executed, enabling the evaluation of different modelling approaches and their impact on network performance.

OpenDSS Reference Simulations

In parallel, the same network models are simulated in OpenDSS to provide reference results. These simulations include both balanced and unbalanced analyses, producing voltage profiles and loading conditions for all network elements. The OpenDSS results serve as a benchmark for validating the accuracy of the pandapower implementation.

Metrics and Validation

Finally, a set of quantitative metrics is computed to compare the results obtained from pandapower and OpenDSS. These include voltage magnitude deviations at the bus level and loading differences for lines and transformers. Both local (per-element) and global metrics are evaluated, allowing for a comprehensive assessment of the consistency and accuracy of the conversion process.

C. Modelling Assumptions and Component Representation

Buses

Every unique bus name encountered in the DSS line definitions is created as a pandapower bus. MV buses are assigned a nominal voltage of 22 kV; the slack (source) bus is assigned 66 kV. LV busbars are assigned 0.4 kV (line-to-line equivalent). Geographic coordinates are attached to all MV buses. For LV busbars that lack explicit coordinates, the position of the upstream MV bus is propagated so that the network remains plottable. Bus names in OpenDSS include phase node suffixes. These suffixes are stripped during parsing so that each physical bus is represented by exactly one pandapower node, regardless of phase configuration.

External Grid

The transmission grid connection is modelled with a fixed voltage setpoint of 1.04 p.u., enabling voltage regulation at the network entry point.
In the OpenDSS model, the source is defined with pu = 1.0, and the difference reflects the use of voltage control mechanisms in pandapower to better match realistic operating conditions.

MV Lines

Medium-voltage lines are modelled by mapping each network connection to the corresponding pandapower line representation, using predefined electrical parameters. Line lengths are consistently normalised to kilometres to ensure uniformity across the model.

Very short connections (<0.001m), corresponding to negligible physical length, are not treated as actual lines but are instead represented as direct bus-to-bus connections. These are modelled as closed switches, while the original line elements are deactivated. This approach improves numerical stability and avoids issues caused by near-zero impedance values in the network model.

LineCodes

OpenDSS Linecode definitions are parsed from OpenDss exports and registered as pandapower line standard types. The following parameter mapping is applied:

Description	OpenDSS parameter	pandapower parameter	Conversion
Positive-sequence resistance per unit length	r_1 [Ω/unit]	r_ohm_per_km	divided by unit-to-km factor
Positive-sequence reactance per unit length	x_1 [Ω/unit]	x_ohm_per_km	divided by unit-to-km factor
Positive-sequence shunt susceptance per unit length	b_1 [μS/unit]	c_nf_per_km	$b_1 \cdot 10^{-6}$ / $(2\pi \cdot 50 \cdot \text{km_factor}) \cdot 10^9$
Maximum allowable current (thermal limit)	normamp [A]	max_i_ka	divided by 1000

HV/MV Substation Transformers

The grid entry transformer steps voltage down from 66 kV to 22 kV and is modelled as a 2-winding transformer. Parameters are set as follows:

Parameter	Value	Description
sn_mva	100MVA	Rated apparent power
vn_hv_kv	66 kV	Nominal voltage at the high-voltage side
vn_lv_kv	22 kV	Nominal voltage at the low-voltage side
vk_percent	5%	Short-circuit voltage, representing the total impedance magnitude
vkr_percent	1%	Copper losses
shift_degree	30°	Phase shift between HV and LV sides due to transformer connection (e.g., Dyn)

Voltage Regulators

Voltage regulators in OpenDSS are modelled as autotransformers using a Jumper–Transformer–Regcontrol arrangement: a series of small single-phase reactors (jumpers) connect the upstream and downstream MV buses, and a per-phase series transformer provides the tap adjustment. In pandapower, each 3-phase regulator set (phases A, B, C) is collapsed into a single 2-winding MV/MV transformer with tap capability. The upstream and downstream MV buses are identified by tracing the jumper reactor connections. The tap-changing behaviour is represented using 17 discrete tap positions, ranging from −8 to +8, with a neutral position at 0. Each tap step corresponds to a voltage change of 1.25%, enabling fine-grained control of the voltage level. The initial tap position is derived from the OpenDSS configuration and mapped to the corresponding discrete position in pandapower. Voltage control is implemented using a discrete tap controller, with a target voltage setpoint of 1.04 pu. This allows the regulator to automatically adjust its tap position in order to maintain the desired voltage level within the network.

MV Shunt Capacitors

Shunt capacitors are modelled as shunt elements connected to the corresponding medium-voltage buses, providing reactive power support to the network. The reactive power values are directly derived from the original model and converted to the appropriate units for pandapower. During time-series simulations, a predefined switching schedule is applied to the capacitor banks. Specifically, capacitors are switched on between 10:00 and 23:00 and remain disconnected outside this time window, approximating typical daytime voltage support operation in distribution networks. This dynamic behaviour allows the assessment of their impact on voltage regulation under varying load conditions [1].

LV Lines

LV lines are read from DSS files and mapped to pandapower lines using the same linecode standard types registered during the MV network build. The LV busbar voltage (vn_kv) is inferred from the transformer LV bus already present in the network. Length unit conversion supports the same set of units as the MV lines.

Loads

Customer loads are not defined as fixed elements in the pandapower network model. Instead, they are incorporated dynamically during the time-series simulation based on the corresponding OpenDSS definitions and load profiles. Each load is assigned to its respective LV bus, with base active and reactive power values derived from the original model parameters.
During the simulation, time-dependent load profiles are applied to scale the power demand at each time step, ensuring consistency with the reference OpenDSS behaviour and enabling an accurate representation of temporal load variations.

MV/LV Distribution Transformers

Low-voltage distribution transformers are modelled using two-winding transformer representations in pandapower. All transformer configurations from the original model — including three-phase Dyn, three-winding split-phase (L–L), and two-winding Yy (pole-top) transformers — are consistently mapped to a unified two-winding model, enabling both balanced and three-phase power flow analysis. The nominal low-voltage level is derived from the secondary voltage value of each

transformer. For three-phase transformers, this value is already provided as a line-to-line voltage and is used directly. For single-phase transformers, the given line-to-neutral voltage is converted to an equivalent line-to-line value by multiplying by √3, ensuring compatibility with the modelling framework. Transformers originating from three-winding (split-phase) configurations are additionally marked with a dedicated flag, allowing their correct treatment in the three-phase model. This enables the assignment of appropriate zero-sequence parameters and improves the representation of asymmetrical behaviour in the network.
The parameter mapping between the original model and pandapower is summarized in the following table:

OpenDSS Parameter	Pandapower Parameter	Description
xhl [%]	vk_percent	Short-circuit voltage magnitude (total impedance)
%loadloss [%]	vkr_percent	Copper losses
%noloadloss [%]	pfe_kw	Core losses, calculated as (%noloadloss/100) · rated power
%imag [%]	i0_percent = 0.1%	Magnetising current; default value used if not specified
wdg1_tap	tap_pos	Tap position mapped to a 5-step discrete tap (positions 1–5, neutral at 3, step 2.5%)
conns = [Delta Wye]	shift_degree = 30°	Phase shift for Dyn transformer configuration

Nominal Power Adjustment for Single-Phase Transformers

For single-phase transformers connected to the 22 kV primary feeder, the nominal power is converted to a three-phase equivalent base by applying a factor of three (×3). This ensures consistency with pandapower's three-phase modelling framework.
In contrast, transformers connected to 12.7 kV (line-to-neutral) branches retain their original nominal rating, as they already represent single-phase sections of the network and do not require scaling.

Isolation Transformers (SWER Transformers)

1. Description

Isolation transformers (ISO transformers) are single-phase medium-voltage (MV/MV) transformers used in rural distribution networks. Their primary function is to electrically isolate downstream MV sub-feeder sections from the main feeder, while enabling the transfer of power between different phase pairs. This allows a portion of the network to be supplied through an alternative phase

configuration, improving flexibility in network operation.
In the original modelling framework, these transformers are defined as three-winding, single-phase units with a Delta–Wye–Wye configuration. They operate entirely at the MV level, with nominal voltage levels corresponding to 22 kV on the primary side and approximately 12.7 kV on the secondary windings. Their rated power is relatively small, typically in the range of 100–200 kVA. The primary winding is connected across two phases of the upstream MV bus, while the secondary windings form a split-phase configuration at the downstream bus, effectively redistributing power into a new feeder section.

1. Modelling Approach

Due to structural differences between the original model and the pandapower framework, these transformers cannot be directly represented as three-winding elements. Specifically, pandapower assumes a hierarchical voltage structure (HV/MV/LV), whereas ISO transformers operate at the same voltage level on both sides.
For this reason, each ISO transformer is approximated using a two-winding MV/MV transformer with equal nominal voltages on both sides (22 kV / 22 kV). This results in a near-unity transformation ratio, where the primary role of the element is to introduce the appropriate series impedance into the network model.
The parameter mapping is summarized in the following table:

OpenDSS Parameter	Pandapower Parameter	Description
vn_hv_kv = vn_lv_kv	22 kV	Equal nominal voltage on both sides (MV/MV)
shift_degree	0°	No phase shift between terminals
i0_percent	0.1	Default magnetising current (not provided in data)

This modelling approach preserves the electrical behaviour of the transformer in terms of impedance, while simplifying the phase-specific characteristics of the original configuration. As a result, the phase-pair reconfiguration present in the original model is not explicitly represented, but its impact on power flow is sufficiently approximated.

2. Representation in Three-Phase Analysis

In the three-phase model, ISO transformers are treated consistently with other MV/MV transformer equivalents and assigned a standard vector group representation. Since the original phase-specific connections cannot be directly modelled, a simplified symmetrical transformer representation is adopted.
Appropriate zero-sequence parameters are assigned to ensure numerical stability and physically meaningful behaviour in the three-phase power flow. This simplification is necessary due to the limitations of the modelling framework, which requires balanced transformer representations, and is

considered acceptable for system-level analysis where detailed phase coupling is not the primary focus.

3. Conclusion

Although ISO transformers primarily serve phase reconfiguration and feeder isolation purposes in the original model, their representation as equivalent MV/MV transformers in pandapower focuses on preserving their electrical impact on power flow, while abstracting away phase-specific behaviour.

D.Simulation Setup

Two parallel simulation pipelines are executed for each network: one in OpenDSS (reference) and one in pandapower (under comparison). Each pipeline is run in both a balanced (positive-sequence) and an unbalanced (3-phase) configuration, yielding four simulation runs per network in total. All runs share the same network topology, load profiles, and selected day.

The selected simulation day corresponds to 15 January (summer period), which represents a high-demand operating condition for the examined networks. This day was chosen as it captures peak loading behaviour, allowing the evaluation of the models under more stressed and realistic operating conditions.

Balanced Simulation

In the balanced simulation, pandapower performs time-series power flow analysis using a positive-sequence network representation. Load demand is modelled through time-dependent profiles, which are applied uniformly across all phases at each time step. The same daily load profiles are used across all simulation tools, ensuring a consistent and comparable input dataset.

Voltage regulation is implemented via discrete tap controllers on regulator transformers, maintaining the voltage close to a predefined setpoint, while shunt capacitors operate according to a daily switching schedule to provide reactive power support.

The simulation is carried out over 48 time steps (30-minute resolution), and key quantities such as bus voltages and element loading are recorded. Results are expressed in per-unit or percentage form, enabling direct comparison with the reference model.

Unbalanced Simulation

In the unbalanced case, pandapower performs full three-phase time-series simulations, explicitly modelling phase asymmetry in the network. Loads are represented using asymmetric elements, where power is distributed across phases according to their actual phase connections.

Three-phase loads are equally distributed among all phases, while single-phase loads are assigned entirely to their respective phase. Each phase component is independently scaled at every time step using the same daily load profiles applied in the balanced case, ensuring consistency in the input data.

The simulation is executed over 48 time steps (30-minute resolution), and at each step a three-phase power flow is solved. Phase-specific results, including bus voltages and loading of lines and transformers, are recorded for all phases, enabling detailed comparison with the OpenDSS reference.

OpenDSS Simulation

In the balanced reference case, OpenDSS performs a full three-phase time-series simulation using the same daily load profiles as the pandapower model, ensuring consistency in the input data.

Although the simulation is inherently unbalanced and phase-resolved, the results are post-processed

to obtain a single representative value per element. Specifically, the mean value across the three phases is calculated for each quantity, such as voltage magnitude and loading.
The simulation is carried out over 48 time steps (30-minute resolution), and the averaged results are expressed in per-unit or percentage form, allowing direct comparison with the balanced pandapower results.

E. Metrics and Comparison Methodology

The comparison between pandapower and OpenDSS is performed using a set of dedicated scripts, each targeting a specific electrical quantity. All scripts follow a common three-step procedure: element matching, metric computation over the full time series, and aggregation into global statistics.

Element Matching

A key challenge in the comparison process is the different identification schemes used by the two tools: pandapower uses integer indices, while OpenDSS uses string-based names.
Matching is performed using the element names stored in the pandapower model. Only elements present in both datasets are included in the comparison. For the three-phase case, phase information is extracted from the naming convention of the OpenDSS outputs and mapped to the corresponding phase-specific results of pandapower.

1. Voltage Conversion to Per-Unit

Since OpenDSS reports voltages in absolute values (Volts) and pandapower in per-unit, a conversion is required. OpenDSS voltages are normalised using the line-to-neutral base voltage of each bus:

$$V_{pu} = \frac{V_{DSS}}{V_{base}}$$

For the balanced case, mean voltage values (averaged across phases) are used and normalised according to the corresponding voltage level (66 kV, 22 kV, or 400 V).

2. Metric Definitions

Metrics are computed over the full time series (48 steps) for each matched element.

2.1 Voltage Metrics (Balanced & 3-Phase)

Errors are expressed as relative percentages:

Metric	Description	Equation
MAPE	Average relative error	$\frac{100}{N} \cdot \sum_{i=1}^{N} \left\lvert \frac{V_{pp,i} - V_{dss,i}}{V_{dss,i}} \right\rvert$
Max error	Maximum deviation	$max\left(\left\lvert \frac{V_{pp,i} - V_{dss,i}}{V_{dss,i}} \right\rvert \right)$
Bias	Mean signed error (over/underestimation)	$\frac{100}{N} \cdot \sum_{i=1}^{N} \left(\frac{V_{pp,i} - V_{dss,i}}{V_{dss,i}} \right)$

2.2 Loading Metrics

Errors are expressed in absolute percentage points:

Metric	Description	Equation
MAE	Mean absolute error	$\frac{1}{N} \cdot \sum_{i=1}^{N} \lvert L_{pp,i} - L_{dss,i} \rvert$
RMSE	Root mean square error	$\sqrt{\frac{1}{N} \cdot \sum_{i=1}^{N} (L_{pp,i} - L_{dss,i})^2}$
MAXAE	Maximum absolute error	$max(\lvert L_{pp,i} - L_{dss,i} \rvert)$
MBE	Mean bias error	$\frac{1}{N} \cdot \sum_{i=1}^{N} (L_{pp,i} - L_{dss,i})$

3. Transformer Exclusion Rules

Certain transformer types are excluded from the unbalanced comparison due to modelling inconsistencies between the two tools:

Category	Reason
ISO Transformers	Not correctly represented in 3-phase pandapower model
1-phase Transformers	Lead to unrealistic loading values in pandapower

Additionally, a correction factor of $\sqrt{3}$ is applied to the OpenDSS current base for single-phase transformers to ensure consistency with pandapower definitions.
The exclusion of these elements ensures a fair and meaningful comparison, focusing only on components that are consistently represented in both simulation environments.

4. *Outputs*

The metrics computation produces both per-element and global summary results, including:

- Per-element error metrics (bus, line, transformer, per phase)
- Global statistics (MAPE, MAE, RMSE, etc.)
- Identification of worst-performing elements

These outputs provide a comprehensive assessment of the agreement between pandapower and OpenDSS across all simulated conditions.

F. Results

The four test networks are real Australian rural and urban distribution feeders at 22 kV, supplied from a 66/22 kV substation. Their key characteristics are summarized below.

Element	Net 1-Rural SMR8	Net 2-Rural KLO14	Net 3-Urban HPK11	Net 4-Urban CRE21
MV buses	3625	3833	595	644
LV buses	8042	10130	10594	6845
MV lines	3610	3823	595	648
LV lines	7338	9430	10551	6766
LV busbars	703	700	45	79
MV/LV 3ph transformers	205	412	45	79
MV/LV 1ph transformers	467	278	0	0
ISO transformers	10	7	0	0
Regulators	2	3	0	0
Capacitors	2	2	0	0
Customer loads	3670	4716	5276	3384
Switches	909	1339	136	181

The selected networks cover both complex rural feeders and more compact urban systems. This diversity ensures that the proposed modelling approach is evaluated under a wide range of operating and structural conditions.

1.Voltages

1.1: Balanced Model — Bus Voltage (vm_pu)

The balanced positive-sequence comparison covers all buses common to both tools, over all 48 time steps of the selected day.

Network	Matched Busses	Total Points	MAE %	Max AE %	Bias %
Net 1-Rural SMR8	11667	560016	2.51	24.86	0.16
Net 2-Rural KLO14	13963	670224	5.02	17.35	-4.77
Net 3-Urban HPK11	11189	537072	3.60	4.97	-3.16
Net 4-Urban CRE21	7489	359472	3.75	5.22	-3.05

The balanced voltage comparison shows a generally good agreement between pandapower and OpenDSS across all examined networks. The mean absolute error (MAE) remains within a moderate range (2.51%–5.02%), indicating that the overall voltage profiles are consistently reproduced.
Urban networks (Net 3 and Net 4) exhibit lower maximum deviations (below 6%), reflecting their more compact topology and reduced feeder lengths, which lead to more stable voltage conditions. In contrast, rural networks (Net 1 and Net 2) show significantly higher maximum errors, reaching up to 24.86%, likely due to long feeder lines, higher impedance, and the presence of lightly loaded or weakly regulated buses.
The bias values reveal a systematic tendency of pandapower to underestimate voltages in most networks, particularly in Net 2 (−4.77%), while Net 1 shows a near-zero bias, indicating a well-balanced error distribution. This suggests that although local deviations may occur, the overall voltage estimation remains unbiased or only slightly shifted.
Overall, the results confirm that the balanced pandapower model provides a reliable approximation of the OpenDSS reference, with deviations primarily driven by network characteristics rather than systematic modelling errors.

1.2: 3-Phase Model — Bus Voltage (vm_pu per phase)

The 3-phase comparison covers matched (bus, phase) pairs. The global metrics are computed across all phases and all time steps simultaneously.

Net 1-Rural SMR8

Net 1-Rural SMR8	Matched Busses	Total Points	MAE %	Max AE %	Bias %
Global	29833	1431984	4.87	35.52	0.12
Phase A	9944	477328	5.45	9.37	-1.64
Phase B	9944	477328	4.73	7.44	0.65
Phase C	9944	477328	4.24	6.47	1.93

Net 2-Rural KLO14

Net 2-Rural KLO14	Matched Busses	Total Points	MAE %	Max AE %	Bias %
Global	35861	1721328	4.64	76.50	0.32
Phase A	11953	573776	5.92	10.90	-1.42
Phase B	11953	573776	4.10	7.16	0.21
Phase C	11953	573776	3.47	5.17	3.00

Net 3-Urban HPK11

Net 3-Urban HPK11	Matched Busses	Total Points	MAE %	Max AE %	Bias %
Global	33567	1611216	4.36	10.04	4.36
Phase A	11189	537072	4.37	5.92	4.37
Phase B	11189	537072	4.34	5.91	4.34
Phase C	11189	537072	1.37	4.92	4.37

Net 4-Urban CRE21

Net 4-Urban CRE21	Matched Busses	Total Points	MAE %	Max AE %	Bias %
Global	22467	1078416	4.20	10.78	4.20
Phase A	7489	359472	4.19	5.42	4.19
Phase B	7489	359472	4.17	5.28	4.17
Phase C	7489	359472	4.23	5.56	4.23

The three-phase voltage comparison demonstrates a moderate but consistent deviation between pandapower and OpenDSS across all networks. The global MAE values are in the range of 4.2%–4.9%, which are notably higher than those observed in the balanced case, reflecting the increased complexity of unbalanced modelling.

A key observation is the relatively small variation between phase-specific MAE values, indicating that the error is not dominated by a single phase but is instead distributed across all phases. However, slight asymmetries are present, particularly in rural networks, where Phase A tends to exhibit higher deviations compared to Phases B and C. This behaviour can be attributed to uneven load distribution and phase-specific connections in the original network.

The global maximum error values are significantly higher than the phase-level maxima, especially in Net 2 (up to 76.50%). This discrepancy is expected, as the global metric captures isolated worst-case outliers across all buses, phases, and time steps, while phase-level values represent averaged behaviour. Such extreme deviations are typically associated with weakly connected buses, low-voltage nodes, or transient operating conditions.

Bias values remain relatively small in most cases, indicating that no strong systematic overestimation or underestimation is present overall. However, individual phase biases differ in sign, suggesting localised modelling discrepancies between the two tools.

Urban networks (Net 3 and Net 4) exhibit more stable behaviour, with lower maximum errors and more consistent phase agreement, while rural networks (Net 1 and Net 2) show larger deviations due to longer feeders, higher impedance, and increased phase imbalance.

Overall, the results indicate that while pandapower captures the general voltage trends in unbalanced operation, the accuracy is reduced compared to the balanced case, and local discrepancies become more pronounced.

2.Line Loading

2.1: Balanced - MV Lines

Network	Matched Lines	MAE (%)	MBE (%)	Max AE (%)
Net 1-Rural SMR8	2701	1.68	-0.68	42.78
Net 2-Rural KLO14	2484	1.26	-0.54	23.35
Net 3-Urban HPK11	456	0.88	-0.88	4.90
Net 4-Urban CRE21	467	0.50	-0.50	4.68

2.2: Balanced - LV Lines

Network	Matched Lines	MAE (%)	MBE (%)	Max AE (%)
Net 1-Rural SMR8	7338	19.58	-19.58	321.68
Net 2-Rural KLO14	9430	19.84	-19.83	345.81
Net 3-Urban HPK11	10550	24.59	-24.59	406.43
Net 4-Urban CRE21	6766	33.68	-33.68	587.87

The comparison of line loading reveals a clear distinction between MV and LV network performance. For MV lines, the agreement between pandapower and OpenDSS is very strong across all networks. The MAE remains low (0.50%–1.68%), indicating that the loading of medium-voltage feeders is accurately captured. The relatively small bias values (close to zero and slightly negative) suggest a mild systematic underestimation by pandapower, but without significant impact on overall accuracy. Maximum deviations are higher in rural networks, reaching up to 42.78%, which can be attributed to long feeders and localized operating conditions, but remain limited in urban networks.

In contrast, LV line loading exhibits significantly larger deviations. The MAE ranges from 19.58% up to 33.68%, while maximum errors reach extremely high values (over 300%–500%). Additionally, the MBE values are almost equal in magnitude to the MAE and consistently negative, indicating a strong systematic underestimation of line loading in pandapower.

This behaviour suggests a structural modelling discrepancy in the LV network representation,

particularly in the way LV loads and feeders are approximated. Similar limitations have been reported in the literature, where simplified LV representations fail to capture detailed feeder behaviour and phase imbalance [1]. Possible causes include differences in load allocation, phase modelling, or simplifications in transformer modelling that affect downstream loading calculations. The amplification of errors in LV networks is expected due to their sensitivity to voltage variations and high R/X ratios.

Overall, while MV line loading results confirm the validity of the modelling approach, the LV results highlight a key limitation of the current conversion methodology and indicate the need for further refinement in low-voltage modelling.
The near equality between MAE and MBE in LV lines indicates that the error is not random but highly systematic, pointing to a consistent modelling bias rather than isolated discrepancies.

2.3: Unbalanced

Net 1-Rural SMR8

Net 1-Rural SMR8	MAE (%)	RMSE (%)	Max AE (%)	MBE (%)
Global	1.59	1.85	139.81	-0.1
Phase A	1.92	2.31	127.23	0.527
Phase B	1.61	1.87	131.67	-0.167
Phase C	1.23	1.37	139.81	-0.67

Net 2-Rural KLO14

Net 2-Rural KLO14	MAE (%)	RMSE (%)	Max AE (%)	MBE (%)
Global	1.81	2.18	172.15	0.51
Phase A	2.69	3.40	172.15	1.48
Phase B	1.49	1.77	70.86	0.04
Phase C	1.24	1.35	85.30	-1.00

Net 3-Urban HPK11

Net 3-Urban HPK11	MAE (%)	RMSE (%)	Max AE (%)	MBE (%)
Global	2.03	2.22	86.10	-2.03
Phase A	2.07	2.27	86.10	-2.07
Phase B	2.09	2.29	84.80	-2.09
Phase C	1.92	2.10	85.39	-1.92

Net 4-Urban CRE21

Net 4-Urban CRE21	MAE (%)	RMSE (%)	Max AE (%)	MBE (%)
Global	2.45	2.57	58.80	-2.34
Phase A	2.35	2.58	54.65	-2.33
Phase B	2.25	2.46	55.58	-2.24
Phase C	2.45	2.68	58.80	-2.45

The unbalanced line loading results demonstrate a generally improved agreement between pandapower and OpenDSS compared to the balanced LV case, with MAE values remaining relatively low (1.5%–2.5%) across all networks. This indicates that the three-phase modelling approach provides a more accurate representation of load distribution and phase-specific behaviour.

The RMSE values are consistently slightly higher than the MAE, confirming the presence of localized larger deviations, but without significantly affecting the overall accuracy. However, the maximum absolute errors remain very high, reaching up to 172.15%, which suggests the presence of extreme outliers. These are likely associated with lightly loaded lines, phase imbalances, or near-zero loading conditions, where small absolute differences translate into large percentage errors.

Phase-level analysis shows moderate variability, particularly in rural networks, where Phase A tends to exhibit higher deviations. This behaviour reflects the inherent asymmetry of real distribution systems and the sensitivity of unbalanced simulations to phase allocation and load modelling assumptions.

The MBE values remain relatively small across all cases, indicating that the deviations are not strongly biased in one direction. In contrast to the balanced LV results, the errors in the unbalanced case are more symmetrically distributed, suggesting that the three-phase modelling reduces systematic bias and improves consistency between the two tools.
Urban networks (Net 3 and Net 4) show more stable and consistent behaviour, with lower maximum deviations and smaller phase discrepancies, while rural networks exhibit higher variability due to longer feeders and more pronounced phase imbalance.
Overall, the results confirm that unbalanced modelling significantly improves the accuracy of line loading representation in pandapower, although extreme outliers remain and should be interpreted with caution.

3. Transformer Loading

3.1: Balanced

Network	MAE (%)	RMSE (%)	Max AE (%)	MBE (%)
Net 1-Rural SMR8	6.56	8.25	145.99	3.03
Net 2-Rural KLO14	7.83	9.59	111.69	5.43
Net 3-Urban HPK11	5.75	6.69	26.24	-1.66
Net 4-Urban CRE21	5.06	6.49	63.30	-0.40

The balanced transformer loading comparison shows moderate deviations between pandapower and OpenDSS, with MAE values ranging from approximately 5% to 8%. This indicates that transformer loading is captured with acceptable accuracy, although less precisely than MV line loading.
RMSE values are consistently higher than MAE, confirming the presence of larger deviations in specific operating conditions. This is further supported by the high maximum errors, which exceed 100% in rural networks, indicating the existence of significant outliers. These extreme values are typically associated with lightly loaded transformers, where small absolute differences result in large percentage deviations.
A notable observation is the positive bias in rural networks (Net 1 and Net 2), indicating that pandapower tends to overestimate transformer loading in these cases. This behaviour may be attributed to differences in load allocation, transformer modelling assumptions, or the treatment of single-phase and split-phase units.
In contrast, urban networks (Net 3 & 4) exhibit smaller deviations and near-zero or slightly negative bias, suggesting a more balanced and consistent representation between the two tools. Overall, the results indicate that while the transformer loading is reasonably well approximated in the balanced model, significant discrepancies remain in specific cases, particularly in rural networks, highlighting the sensitivity of transformer loading calculations to modelling assumptions and data representation.

3.2: Unbalanced

Net 1-Rural SMR8

Net 1-Rural SMR8	MAE (%)	RMSE (%)	Max AE (%)	MBE (%)
Global	25.7	28.74	1663.62	24.61
Phase A	27.02	30.42	794.52	25.85
Phase B	29.46	32.61	1119.28	28.57
Phase C	20.63	23.2	1663.62	19.47

Net 2-Rural KLO14

Net 2-Rural KLO14	MAE (%)	RMSE (%)	Max AE (%)	MBE (%)
Global	20.68	23.49	1152.13	19.62
Phase A	29.92	34.34	1152.13	29.01
Phase B	13.44	15.38	590.01	12.26
Phase C	18.69	20.75	986.43	17.60

Net 3-Urban HPK11

Net 3-Urban HPK11	MAE (%)	RMSE (%)	Max AE (%)	MBE (%)
Global	4.08	4.59	22.46	2.97
Phase A	3.94	4.44	20.10	2.66
Phase B	4.44	5.00	22.46	3.74
Phase C	3.86	4.34	17.86	2.50

Net 4-Urban CRE21

Net 4-Urban CRE21	MAE (%)	RMSE (%)	Max AE (%)	MBE (%)
Global	91.74	99.06	2285.284	89.22
Phase A	71.38	77.1	1249.91	70.45
Phase B	83.33	89.36	1366.71	82.04
Phase C	120.51	130.73	2285.284	115.19

The unbalanced transformer loading results reveal significantly larger deviations between pandapower and OpenDSS compared to all previous cases. In rural networks, the MAE ranges from approximately 20% to 26%, while in the urban Net 4 case it reaches extremely high values (above 90%), indicating substantial discrepancies in transformer loading estimation.
The RMSE values closely follow the MAE, confirming that the deviations are not limited to isolated outliers but instead represent consistently high errors across multiple operating points. This is further supported by the extremely large maximum errors, which exceed 1000% and reach up to 2285%, indicating severe mismatches in specific transformer elements or time steps.
A key observation is the very high MBE values, which are of similar magnitude to the MAE in all cases. This indicates a strong systematic bias, with pandapower consistently overestimating transformer loading in the unbalanced simulations. Unlike previous results, where errors were more symmetrically distributed, the deviations here are clearly directional.
The discrepancy is particularly pronounced in Net 4, where all phases exhibit extremely high errors. A more detailed analysis shows that these deviations are primarily driven by a subset of very small transformers present in the network, including multiple units rated at 10 kVA and 25 kVA. For these low-capacity transformers, the unbalanced modelling approach leads to disproportionately high loading values in pandapower (up to 500–1200%), while the corresponding OpenDSS values remain within a realistic range (approximately 4–20%).
To further investigate this behaviour, the transformer loading error was analysed as a function of transformer rating, as shown below:

Transformer Rating	MAE (%)
10 kVA	310%
25 kVA	44%
50–100 kVA	19-47%
>200 kVA	3-5%

The results clearly demonstrate that the loading error decreases significantly with increasing transformer rating. Very small transformers (10 kVA) exhibit extremely high errors, while 25 kVA units show reduced but still substantial deviations. As transformer size increases, the error decreases considerably, with transformers rated above 200 kVA showing very good agreement with the reference model.

This trend indicates that the observed discrepancies are not uniformly distributed across the network but are strongly dependent on transformer rating. Low-capacity transformers are particularly sensitive to modelling assumptions and unbalanced loading conditions, where small absolute differences in power translate into disproportionately large percentage errors. In contrast, larger transformers operate within a more stable loading range, resulting in more consistent and reliable behaviour.

These findings suggest that the discrepancies arise from structural modelling limitations rather than random numerical differences. Contributing factors may include the simplified representation of single-phase and split-phase transformers, differences in load allocation across phases, and the use of equivalent two-winding transformer models in pandapower.

In contrast, Net 3 (urban) shows significantly better agreement, with MAE values around 4%, indicating that the issue is not inherent to the methodology but strongly dependent on network characteristics and transformer configuration, particularly the presence of low-capacity units.

Overall, the results highlight a key limitation of the current modelling approach in unbalanced transformer loading. While the model performs adequately for medium and large transformers, it fails to accurately represent the behaviour of small-capacity units under unbalanced conditions. This suggests that further refinement is required, particularly in the representation of phase-specific transformer behaviour and low-voltage network interactions.

4. Overall Comparison Summary

Case	Quantity	MAE (%)	Max AE (%)	Bias (%)	Observation
Balanced	Voltage	3-5	5-25	Small	Good agreement
3-Phase	Voltage	4-5	up to 76	Small	Good agreement
Balanced	MV Line Loading	0.5-1.7	5-40	Small	Excellent agreement
Balanced	LV Line Loading	20-34	300-600	Large(-)	Significant deviation
3-Phase	Line Loading	1.5-2.5	50-170	Small	Good agreement
Balanced	Transformer Loading	5-8	25-150	Medium	Moderate deviation
3-Phase	Transformer Loading	5-90	up to 2000	Large(+)	Significant deviation

The summary table provides a consolidated view of the comparison results across all examined quantities and modelling approaches. The qualitative observation labels are assigned based on the magnitude of the error metrics, primarily MAE and bias values, offering an intuitive assessment of model accuracy.
The results indicate that pandapower achieves excellent to good agreement with OpenDSS in balanced simulations and for medium-voltage (MV) network elements, where modelling assumptions are relatively consistent and system behaviour is more stable. In contrast, low-voltage (LV) networks exhibit significant deviations under balanced conditions, mainly due to simplified modelling assumptions and the aggregated representation of loads.
The introduction of unbalanced (three-phase) modelling leads to a substantial improvement in LV line loading accuracy, reducing systematic bias and providing a more realistic representation of phase-specific behaviour. At the same time, this increased modelling detail reveals important limitations, particularly in transformer loading calculations, where large deviations and strong systematic bias are observed.
These findings highlight that while simplified (balanced) models may provide stable and computationally efficient results, they can mask underlying modelling inaccuracies. Conversely, more detailed unbalanced models improve accuracy in some aspects (e.g., LV lines) but also expose structural limitations in others (e.g., transformer representation).
The analysis demonstrates that model accuracy depends not only on the level of detail but also on the appropriateness of the modelling assumptions, particularly in the representation of LV networks and transformer behaviour.

G. Conclusions

This work presented a systematic framework for the conversion and validation of distribution network models from OpenDSS to pandapower, enabling a comprehensive comparison under both balanced and unbalanced operating conditions.

The results of the balanced simulations demonstrate a generally strong agreement between the two tools, particularly for bus voltage profiles and medium-voltage (MV) line loading. Voltage deviations remain within acceptable limits, and MV line loading is reproduced with high accuracy and minimal bias, confirming the reliability of the conversion methodology for positive-sequence analysis.

However, significant discrepancies are observed in low-voltage (LV) line loading under balanced conditions. The high MAE values, combined with the near equality between MAE and MBE, indicate a strong systematic underestimation by pandapower. This behaviour suggests that the errors are not random but originate from structural modelling differences, including load allocation, voltage base assumptions, and transformer representation. This observation is consistent with previous studies that highlight the limitations of simplified LV modelling approaches in accurately capturing distribution network behaviour [1].

The introduction of unbalanced (three-phase) modelling leads to a notable improvement in LV line loading accuracy. The substantial reduction in bias indicates that explicitly modelling phase asymmetry provides a more realistic representation of load distribution and network behaviour.

Transformer loading results reveal a clear contrast between balanced and unbalanced simulations. In

the balanced case, transformer loading is estimated with moderate accuracy, with MAE values in the range of 5%–8% and relatively limited bias, indicating a reasonably consistent representation between pandapower and OpenDSS.

In contrast, the unbalanced simulations exhibit significantly larger deviations, particularly in rural networks, where MAE values exceed 20% and extreme maximum errors are observed. The close correspondence between MAE and MBE indicates a strong systematic overestimation of transformer loading. This behaviour highlights fundamental modelling limitations, primarily related to the representation of single-phase and split-phase transformers using equivalent two-winding models in pandapower.

Urban networks consistently demonstrate more stable and accurate results across all metrics, while rural networks exhibit larger deviations due to longer feeders, higher impedance, and increased phase imbalance.

Overall, the study confirms that pandapower can reliably reproduce OpenDSS results for balanced simulations and MV network behaviour. However, challenges remain in accurately modelling LV networks and transformer loading under unbalanced conditions. These findings indicate that while the balanced approach provides a stable approximation, the unbalanced modelling reveals underlying structural inconsistencies that require further refinement.

Future work may focus on improving transformer modelling techniques, enhancing LV network representation, and incorporating more detailed phase-specific behaviour in order to achieve greater consistency between the two simulation environments.

H.References

[1] A. T. Procopiou, J. Jazaeri, K. Petrou, M. Liu, L. F. Ochoa, T. Langstaff, J. Harding, and J. Theunissen, "On the Role of Integrated MV-LV Network Modelling in DER Studies",
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